Computational modeling reveals a potential selection bias in high-density surface electromyography analysis of diabetic neuropathy

> Renato Naville Watanabe and Marcos Duarte, Federal University of ABC, Brazil

Dependencies

Execution provenance

Item	Value
Executed at	`2026-08-05T00:43:50-03:00`
Python	`3.12.7`
marimo	`0.23.16`
NumPy	`1.26.0`
pandas	`3.0.5`
Matplotlib	`3.11.1`
SciPy	`1.16.3`
Git commit	`42c244baa8ce058dd28e2ba94973af2eb50097bd`
Git state	`clean`
Notebook SHA-256	`207ae631b768a7e8a3ca7e434c3dc7e253e705266479748a9b3d28c73a4af53f`

Analysis configuration

Analysis helpers

All-active-MU simulation truth

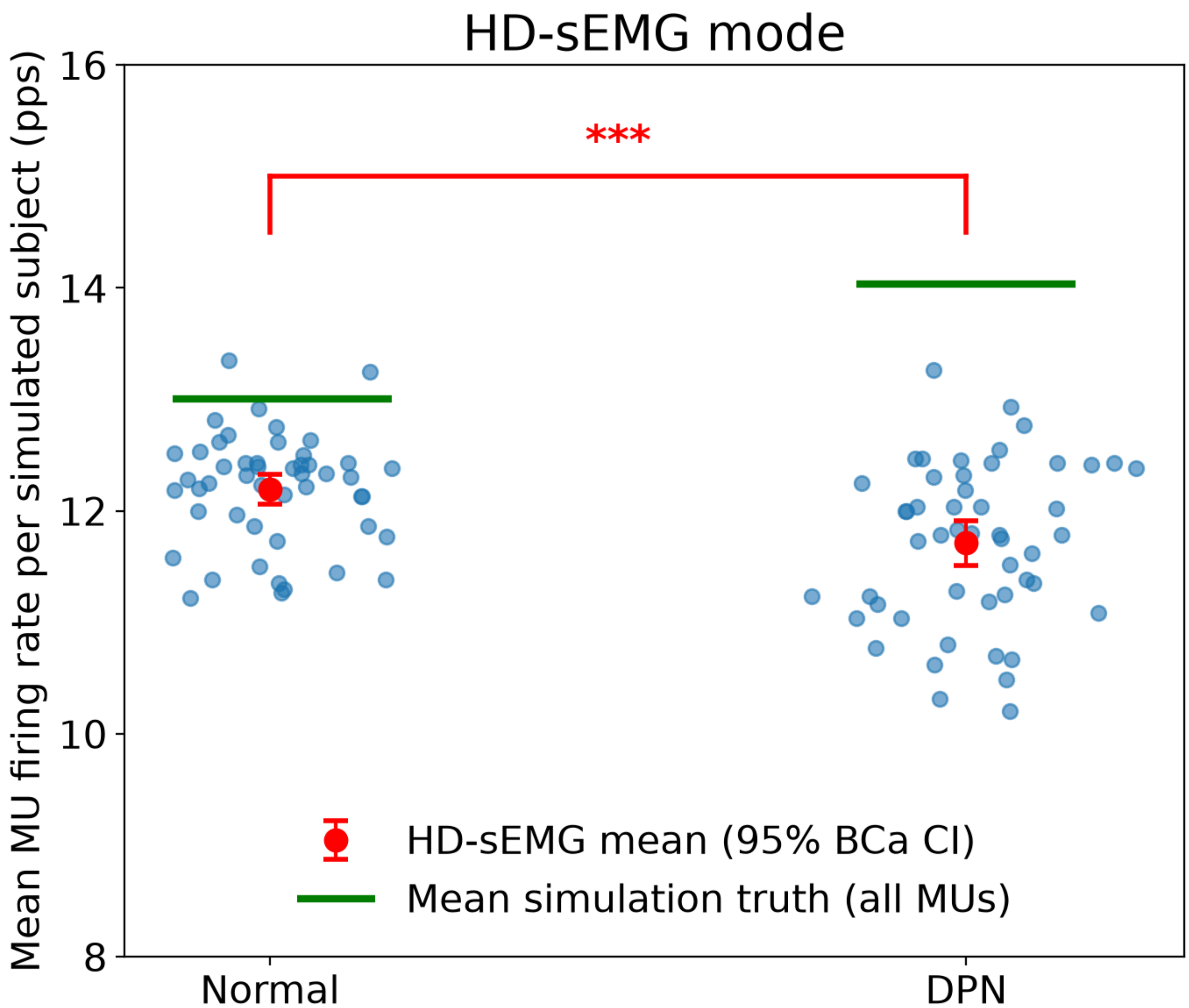
For each simulated subject, the mean firing rate across every active motor unit is known exactly and defines the trial-specific simulation truth. This full-population value is the sole reference for the sampled HD-sEMG estimate.

```
============================================================
=== SIMULATION TRUTH (all motor units) ===
============================================================
FR simulation means (n=50 paired simulations/subjects): normal 13.00 ± 0.27
[95% BCa CI 12.93, 13.08]; DPN 14.04 ± 0.40 [95% BCa CI 13.93, 14.14]
Paired comparison (Simulation truth (all motor units)): DPN - normal = 1.03
pps [95% BCa CI 0.93, 1.13]; Wilcoxon signed-rank W=0.0, p=1.7764e-15
ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.217 ±
0.014 [95% BCa CI 0.213, 0.221]; DPN 0.150 ± 0.015 [95% BCa CI 0.146, 0.155]
Paired ISI-CoV comparison (Simulation truth (all motor units)): DPN - normal
= -0.067 [95% BCa CI -0.071, -0.063]; Wilcoxon signed-rank W=0.0, p=1.7764e-
15
Analyzed MUs per simulation/subject (mean ± SD [range]): normal 174.5 ± 2.9
[170, 182]; DPN 161.8 ± 3.1 [156, 171]
```

Firing rates at 20% MVC

Each simulation represents one subject. Group means, SDs, confidence intervals, and tests for firing rate and ISI-CoV therefore use one mean per simulation; raw motor-unit values are retained only for explicitly descriptive analyses. In the HD-sEMG mode, 10 unique motor units are drawn without replacement from those meeting the firing-rate and ISI-CoV eligibility criteria, using the configured fixed seed. Red horizontal lines show the condition means of the subject-specific simulation truths calculated from all active MUs.

```
============================================================
=== HD-sEMG MODE (seeded random selection) ===
============================================================
Selection criteria: {'fmin': 5, 'fmax': 15, 'isicv': 0.3}
Motor units per simulation: 10
Selection seed: 20260102
FR simulation means (n=50 paired simulations/subjects): normal 12.20 ± 0.50
[95% BCa CI 12.06, 12.33]; DPN 11.71 ± 0.72 [95% BCa CI 11.51, 11.91]
Paired comparison (HD-sEMG): DPN - normal = -0.48 pps [95% BCa CI -0.73, -0.
25]; Wilcoxon signed-rank W=261.0, p=7.9659e-04
ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.153 ±
0.014 [95% BCa CI 0.149, 0.156]; DPN 0.126 ± 0.018 [95% BCa CI 0.121, 0.131]
Paired ISI-CoV comparison (HD-sEMG): DPN - normal = -0.027 [95% BCa CI -0.03
3, -0.020]; Wilcoxon signed-rank W=58.0, p=1.4433e-10
Analyzed MUs per simulation/subject (mean ± SD [range]): normal 10.0 ± 0.0
[10, 10]; DPN 10.0 ± 0.0 [10, 10]
```




```
Primary comparison figure saved to: diabetes/figures/mn_firing_rate_comparison_combined_v2.png
Statistical results saved to: diabetes/csv_results/mn_firing_rate_p_values_combined_v2.csv
```

Force-production statistics

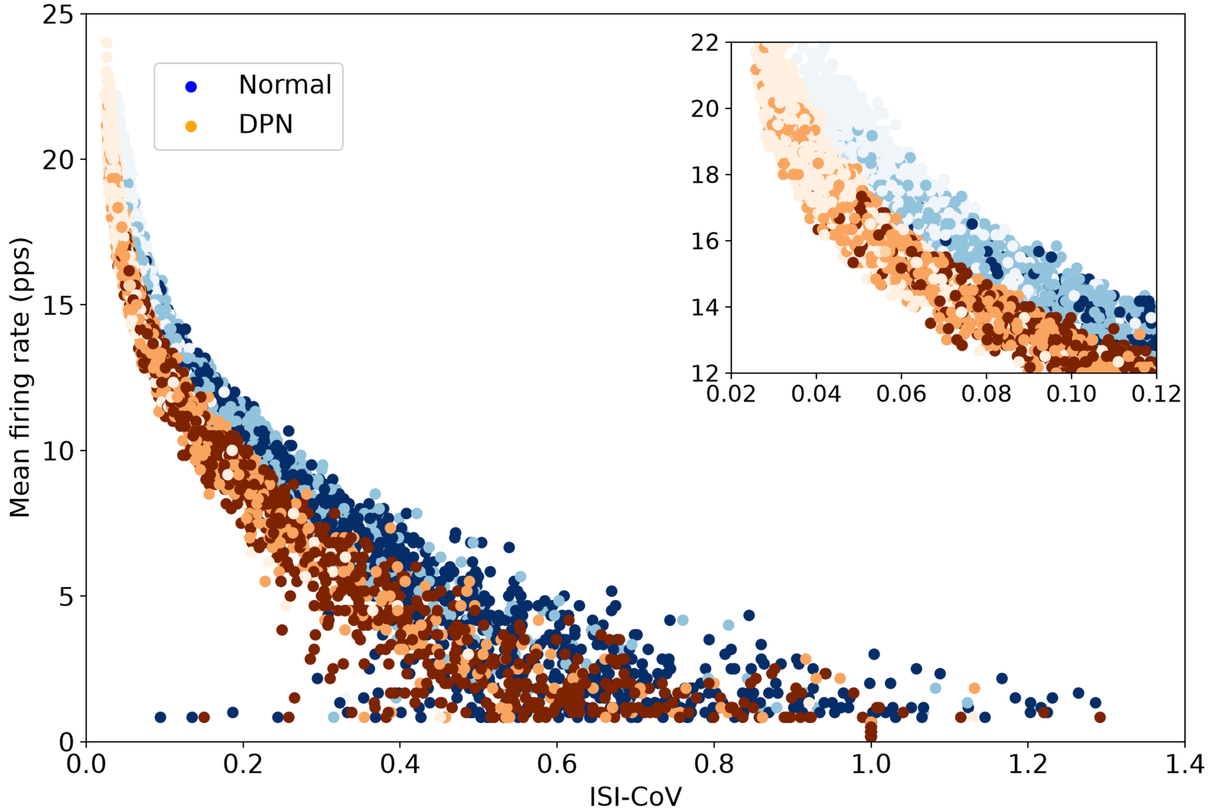
Each simulation represents one subject and contributes one steady-state mean force and force coefficient of variation (CoV) per group. Normal–DPN estimates use paired simulation IDs, with 95% BCa bootstrap intervals and two-sided Wilcoxon signed-rank p-values.

```
============================================================
=== FORCE PRODUCTION STATISTICS (20% MVC, batch: variability) ===
============================================================
(Steady-state: t > 4000 ms)
(Bootstrap resamples: 100,000)

--- normal (n=50 paired simulations/subjects; 50 available) ---
  Mean force: 59.9939 ± 0.0484 [95% BCa CI: 59.9815, 60.0081]
  Force as %MVC: 20.00%
  CoV of force: 0.0212 ± 0.0030 [95% BCa CI: 0.0205, 0.0222]

--- DPN (n=50 paired simulations/subjects; 50 available) ---
  Mean force: 60.0113 ± 0.0572 [95% BCa CI: 59.9945, 60.0259]
  Force as %MVC: 20.00%
  CoV of force: 0.0147 ± 0.0028 [95% BCa CI: 0.0140, 0.0155]

--- Paired comparison (DPN - normal) ---
  Mean-force difference: 0.0173 [95% BCa CI: -0.0034, 0.0360]; paired Wilcox
on W=445.0, p=6.3466e-02
  Force-CoV difference: -0.0065 [95% BCa CI: -0.0075, -0.0055]; paired Wilco
xon W=3.0, p=8.8818e-15

--- Between-simulation/subject variability ---
  normal: CoV of mean force across simulations/subjects = 0.08%
  normal: CoV of force CoV across simulations/subjects = 14.22%
  DPN: CoV of mean force across simulations/subjects = 0.10%
  DPN: CoV of force CoV across simulations/subjects = 18.73%

============================================================
=== FORCE PRODUCTION STATISTICS (10% MVC, batch: mvc10) ===
============================================================
(Steady-state: t > 4000 ms)
(Bootstrap resamples: 100,000)

--- normal (n=50 paired simulations/subjects; 50 available) ---
  Mean force: 30.0396 ± 0.0398 [95% BCa CI: 30.0289, 30.0508]
  Force as %MVC: 10.01%
  CoV of force: 0.0274 ± 0.0048 [95% BCa CI: 0.0261, 0.0287]

--- DPN (n=50 paired simulations/subjects; 50 available) ---
  Mean force: 30.0197 ± 0.0445 [95% BCa CI: 30.0070, 30.0313]
  Force as %MVC: 10.01%
  CoV of force: 0.0182 ± 0.0036 [95% BCa CI: 0.0173, 0.0192]

--- Paired comparison (DPN - normal) ---
  Mean-force difference: -0.0200 [95% BCa CI: -0.0350, -0.0052]; paired Wilc
oxon W=383.0, p=1.3305e-02
  Force-CoV difference: -0.0091 [95% BCa CI: -0.0110, -0.0075]; paired Wilco
xon W=10.0, p=7.6383e-14

--- Between-simulation/subject variability ---
  normal: CoV of mean force across simulations/subjects = 0.13%
  normal: CoV of force CoV across simulations/subjects = 17.60%
  DPN: CoV of mean force across simulations/subjects = 0.15%
  DPN: CoV of force CoV across simulations/subjects = 19.63%
```

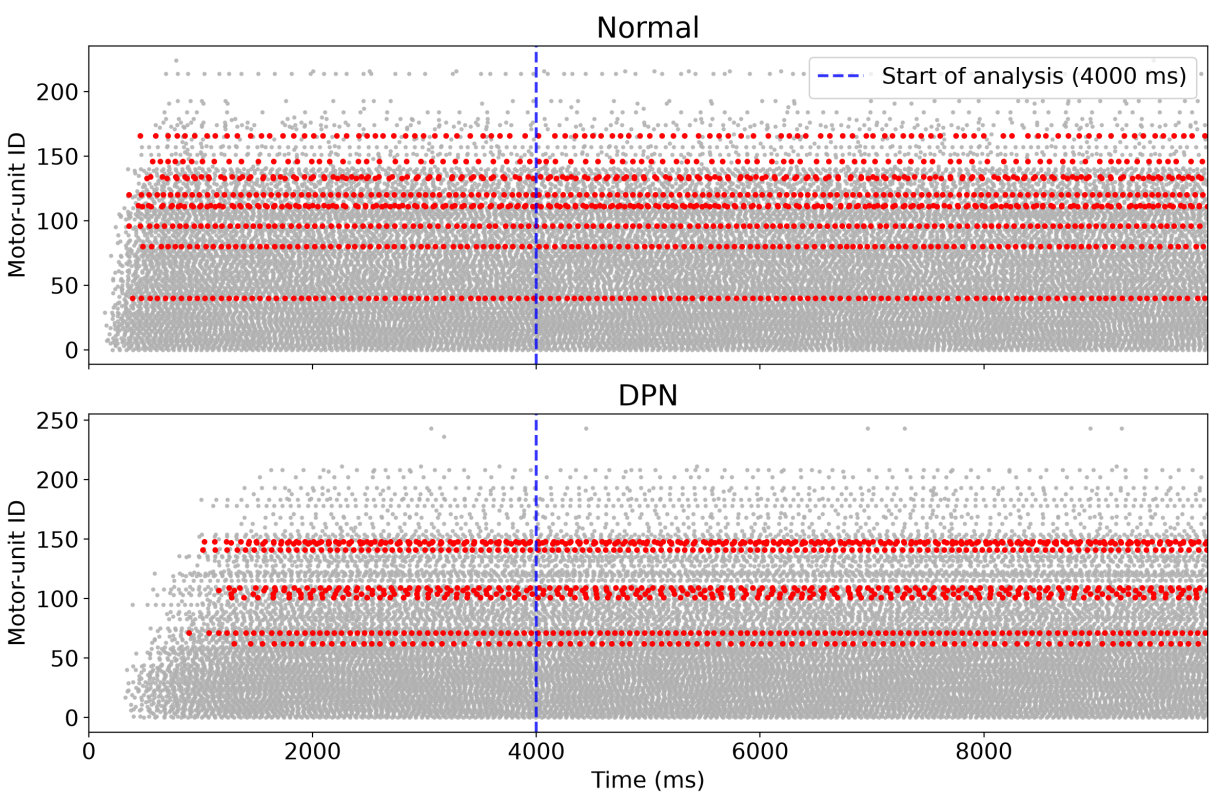
```
============================================================
=== FORCE PRODUCTION STATISTICS (50% MVC, batch: mvc50) ===
============================================================
(Steady-state: t > 4000 ms)
(Bootstrap resamples: 100,000)

--- normal (n=50 paired simulations/subjects; 50 available) ---
  Mean force: 150.0022 ± 0.0788 [95% BCa CI: 149.9796, 150.0230]
  Force as %MVC: 50.00%
  CoV of force: 0.0159 ± 0.0019 [95% BCa CI: 0.0155, 0.0165]

--- DPN (n=50 paired simulations/subjects; 50 available) ---
  Mean force: 150.0128 ± 0.0706 [95% BCa CI: 149.9948, 150.0336]
  Force as %MVC: 50.00%
  CoV of force: 0.0106 ± 0.0017 [95% BCa CI: 0.0101, 0.0110]

--- Paired comparison (DPN - normal) ---
  Mean-force difference: 0.0106 [95% BCa CI: -0.0181, 0.0419]; paired Wilcox
on W=598.0, p=7.0895e-01
  Force-CoV difference: -0.0054 [95% BCa CI: -0.0060, -0.0048]; paired Wilco
xon W=0.0, p=1.7764e-15

--- Between-simulation/subject variability ---
  normal: CoV of mean force across simulations/subjects = 0.05%
  normal: CoV of force CoV across simulations/subjects = 11.67%
  DPN: CoV of mean force across simulations/subjects = 0.05%
  DPN: CoV of force CoV across simulations/subjects = 16.50%
```

Firing-rate and ISI-variability relationship

This scatterplot is an explicitly descriptive motor-unit-level view; it is not used for group-level inference.



Motor-unit selection visualizations

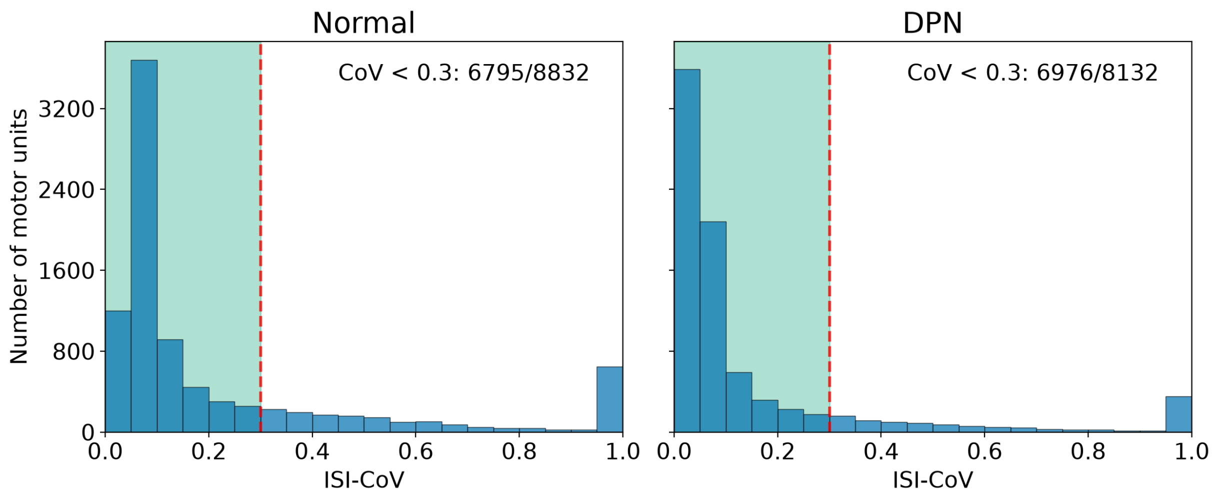



```
Motor-unit selection figure saved to: diabetes/figures/what_mn_selected_comb
ined_trial_30_v2.png
```

```
Condition normal: min index=6, max index=210; mean ID span=107.6 ± 22.2
Condition DPN: min index=4, max index=247; mean ID span=127.4 ± 34.8
Paired subjects with larger DPN ID span: 33/50
Selected-MU ID summary saved to: diabetes/csv_results/hdsemg_selected_motor_units_v2.csv
```

ISI-variability distributions

These histograms describe individual motor units pooled across simulations; they are not used for subject-level estimates or tests.




```
ISI CoV histograms for all motor units saved to: diabetes/figures/isi_cov_histograms_all_units_variability_v2.png
```

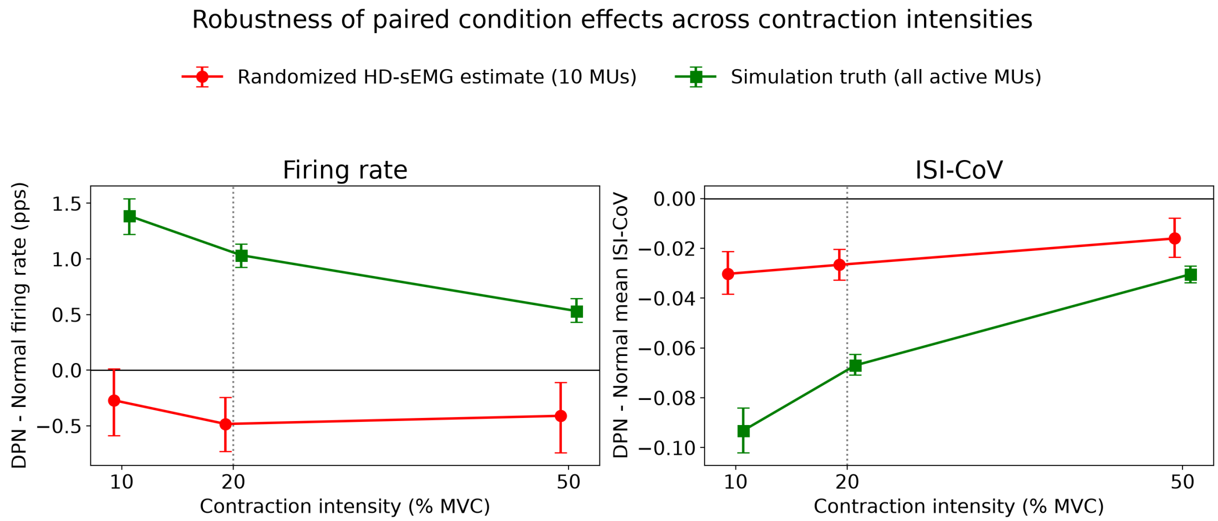
Robustness across contraction intensities

The 20% MVC condition was designated as the primary reference analysis. The same simulation-level analysis was repeated at 10% and 50% MVC using 50 paired simulated subjects per force level to assess whether the direction and magnitude of the findings were robust across contraction intensities.

For firing rate and ISI-CoV, the analysis estimates the paired DPN-minus-Normal effect at each MVC, quantifies the difference between the randomized HD-sEMG estimate and the all-active-MU simulation truth, and tests force-level dependence using a within-subject Friedman test followed by all three paired MVC contrasts with Holm adjustment. Trial identifiers must match within each condition pair and across contraction intensities.

```
============================================================
=== FIRING RATES AT 10% MVC (batch: mvc10) ===
============================================================

--- Randomized HD-sEMG selection ---
Selection criteria: {'fmin': 5, 'fmax': 15, 'isicv': 0.3}
Selection seed: 20260104
FR simulation means (n=50 paired simulations/subjects): normal 12.03 ± 0.66
[95% BCa CI 11.85, 12.21]; DPN 11.76 ± 0.80 [95% BCa CI 11.54, 11.98]
Paired comparison (10% MVC HD-sEMG): DPN - normal = -0.27 pps [95% BCa CI -
0.59, 0.01]; Wilcoxon signed-rank W=479.5, p=1.8581e-01
ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.167 ±
0.020 [95% BCa CI 0.162, 0.172]; DPN 0.137 ± 0.023 [95% BCa CI 0.131, 0.143]
Paired ISI-CoV comparison (10% MVC HD-sEMG): DPN - normal = -0.030 [95% BCa
CI -0.038, -0.021]; Wilcoxon signed-rank W=94.0, p=5.3101e-09
Analyzed MUs per simulation/subject (mean ± SD [range]): normal 10.0 ± 0.0
[10, 10]; DPN 10.0 ± 0.0 [10, 10]

--- Simulation truth (all active MUs) ---
FR simulation means (n=50 paired simulations/subjects): normal 10.13 ± 0.39
[95% BCa CI 10.02, 10.24]; DPN 11.52 ± 0.44 [95% BCa CI 11.39, 11.64]
Paired comparison (10% MVC all motor units): DPN - normal = 1.39 pps [95% BC
a CI 1.22, 1.54]; Wilcoxon signed-rank W=0.0, p=1.7764e-15
ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.321 ±
0.021 [95% BCa CI 0.315, 0.327]; DPN 0.228 ± 0.023 [95% BCa CI 0.221, 0.234]
Paired ISI-CoV comparison (10% MVC all motor units): DPN - normal = -0.093
[95% BCa CI -0.102, -0.084]; Wilcoxon signed-rank W=0.0, p=1.7764e-15
Analyzed MUs per simulation/subject (mean ± SD [range]): normal 138.6 ± 3.7
[129, 147]; DPN 123.3 ± 3.0 [118, 131]

============================================================
=== FIRING RATES AT 50% MVC (batch: mvc50) ===
============================================================

--- Randomized HD-sEMG selection ---
Selection criteria: {'fmin': 5, 'fmax': 15, 'isicv': 0.3}
Selection seed: 20260105
FR simulation means (n=50 paired simulations/subjects): normal 11.90 ± 0.72
[95% BCa CI 11.71, 12.10]; DPN 11.49 ± 0.90 [95% BCa CI 11.23, 11.73]
Paired comparison (50% MVC HD-sEMG): DPN - normal = -0.41 pps [95% BCa CI -
0.74, -0.11]; Wilcoxon signed-rank W=424.0, p=3.9299e-02
ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.141 ±
0.019 [95% BCa CI 0.136, 0.147]; DPN 0.125 ± 0.021 [95% BCa CI 0.120, 0.131]
Paired ISI-CoV comparison (50% MVC HD-sEMG): DPN - normal = -0.016 [95% BCa
CI -0.024, -0.008]; Wilcoxon signed-rank W=272.0, p=2.7386e-04
Analyzed MUs per simulation/subject (mean ± SD [range]): normal 10.0 ± 0.0
[10, 10]; DPN 10.0 ± 0.0 [10, 10]

--- Simulation truth (all active MUs) ---
FR simulation means (n=50 paired simulations/subjects): normal 16.88 ± 0.23
[95% BCa CI 16.82, 16.95]; DPN 17.42 ± 0.38 [95% BCa CI 17.31, 17.52]
Paired comparison (50% MVC all motor units): DPN - normal = 0.53 pps [95% BC
a CI 0.43, 0.64]; Wilcoxon signed-rank W=6.0, p=2.4869e-14
ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.115 ±
0.008 [95% BCa CI 0.113, 0.117]; DPN 0.085 ± 0.010 [95% BCa CI 0.082, 0.087]
```

```
Paired ISI-CoV comparison (50% MVC all motor units): DPN - normal = -0.030
[95% BCa CI -0.034, -0.027]; Wilcoxon signed-rank W=0.0, p=1.7764e-15
Analyzed MUs per simulation/subject (mean ± SD [range]): normal 228.4 ± 2.1
[223, 233]; DPN 221.2 ± 2.4 [215, 225]

============================================================================
==
=== FORCE-LEVEL ROBUSTNESS OF DPN - NORMAL EFFECTS ===
============================================================================
==

firing_rate (pps)
  10% MVC, randomized_hdsemg: -0.2707 [-0.5880, +0.0143], p=0.1858, Holm p=
0.1858
  20% MVC, randomized_hdsemg: -0.4840 [-0.7320, -0.2450], p=0.0007966, Holm
p=0.00239
  50% MVC, randomized_hdsemg: -0.4107 [-0.7407, -0.1100], p=0.0393, Holm p=
0.0786
  10% MVC, simulation_truth_all_active_motor_units: +1.3862 [+1.2203, +1.543
0], p=1.776e-15, Holm p=5.329e-15
  20% MVC, simulation_truth_all_active_motor_units: +1.0331 [+0.9252, +1.133
4], p=1.776e-15, Holm p=5.329e-15
  50% MVC, simulation_truth_all_active_motor_units: +0.5314 [+0.4294, +0.642
8], p=2.487e-14, Holm p=2.487e-14
  MVC dependence, randomized_hdsemg: Friedman Q=4.120, p=0.1275
    10_minus_20: +0.2133 [-0.2013, +0.5953], Holm p=0.5775
    50_minus_20: +0.0733 [-0.3460, +0.4923], Holm p=1
    50_minus_10: -0.1400 [-0.5517, +0.2603], Holm p=1
  MVC dependence, simulation_truth_all_active_motor_units: Friedman Q=36.28
0, p=1.324e-08
    10_minus_20: +0.3530 [+0.1781, +0.5113], Holm p=0.0001702
    50_minus_20: -0.5018 [-0.6429, -0.3455], Holm p=1.681e-07
    50_minus_10: -0.8548 [-1.0601, -0.6361], Holm p=8.439e-09

isi_cv (unitless)
  10% MVC, randomized_hdsemg: -0.0302 [-0.0383, -0.0213], p=5.31e-09, Holm p
=1.062e-08
  20% MVC, randomized_hdsemg: -0.0266 [-0.0327, -0.0204], p=1.443e-10, Holm
p=4.33e-10
  50% MVC, randomized_hdsemg: -0.0160 [-0.0236, -0.0079], p=0.0002739, Holm
p=0.0002739
  10% MVC, simulation_truth_all_active_motor_units: -0.0931 [-0.1020, -0.084
0], p=1.776e-15, Holm p=5.329e-15
  20% MVC, simulation_truth_all_active_motor_units: -0.0670 [-0.0709, -0.062
6], p=1.776e-15, Holm p=5.329e-15
  50% MVC, simulation_truth_all_active_motor_units: -0.0304 [-0.0338, -0.027
1], p=1.776e-15, Holm p=5.329e-15
  MVC dependence, randomized_hdsemg: Friedman Q=5.880, p=0.05287
    10_minus_20: -0.0036 [-0.0141, +0.0078], Holm p=0.323
    50_minus_20: +0.0106 [+0.0006, +0.0202], Holm p=0.09662
    50_minus_10: +0.0142 [+0.0033, +0.0255], Holm p=0.04594
  MVC dependence, simulation_truth_all_active_motor_units: Friedman Q=61.72
0, p=3.96e-14
    10_minus_20: -0.0261 [-0.0363, -0.0158], Holm p=1.976e-05
    50_minus_20: +0.0366 [+0.0313, +0.0416], Holm p=2.665e-14
    50_minus_10: +0.0627 [+0.0522, +0.0723], Holm p=4.867e-13
```




```
Force-level robustness figure saved to: diabetes/figures/force_level_robustn
ess_v2.png
Statistical summaries saved to diabetes/csv_results/force_level_*_v2.csv
```

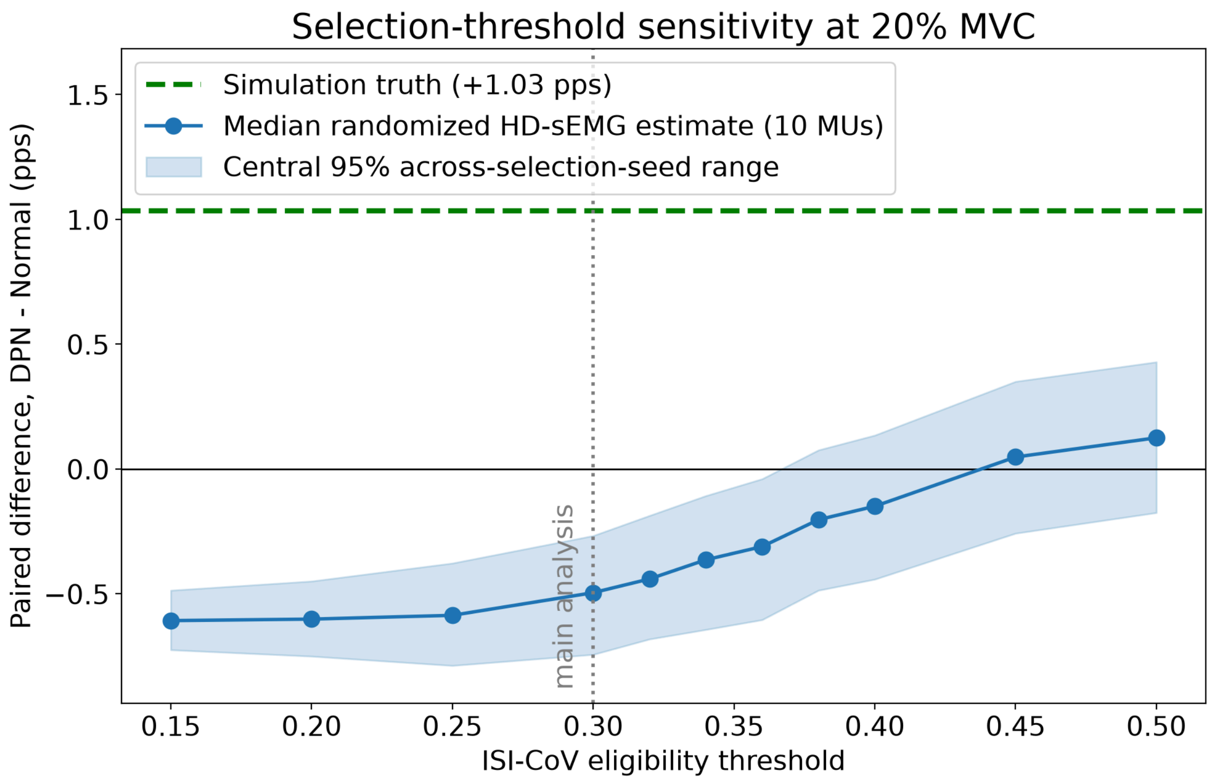
Selection-threshold sensitivity analysis

At 20% MVC, the ISI-CoV eligibility threshold was varied while holding the remaining selection criteria constant to assess the sensitivity of HD-sEMG-like estimates to this analytical choice.

The HD-sEMG analysis first applies a firing-rate window (`fmin < FR < fmax`) and an ISI-CoV ceiling, then randomly samples 10 eligible motor units without replacement. Per-MU firing rates and ISI-CoV values are computed once per simulated subject and reused across all analyses below.

Selection stability is evaluated over 1,000 prespecified seeds at 20%, 10%, and 50% MVC. The threshold analysis uses that same ordered seed set at every ISI-CoV threshold (common random numbers), so differences along the curve are less affected by unrelated draw noise. Its shaded band is the central 95% across-selection-seed range, not a confidence interval. The known mean over all active motor units is shown separately as the simulation truth.

```
============================================================================
==
=== RANDOMIZED HD-sEMG SELECTION-STABILITY ANALYSIS ===
============================================================================
==
20% MVC (50 pairs): fixed seed 20260102, difference -0.484 pps, p=0.0007966
   Across-selection-seed range: -0.743 to -0.267 pps; median -0.495
   Eligible pool (minimum/median/maximum): 32/46.0/61
   Simulation truth: +1.033 pps, p=1.776e-15
10% MVC (50 pairs): fixed seed 20260104, difference -0.271 pps, p=0.1858
   Across-selection-seed range: -0.630 to -0.163 pps; median -0.398
   Eligible pool (minimum/median/maximum): 34/49.0/65
   Simulation truth: +1.386 pps, p=1.776e-15
50% MVC (50 pairs): fixed seed 20260105, difference -0.411 pps, p=0.0393
   Across-selection-seed range: -0.472 to +0.064 pps; median -0.205
   Eligible pool (minimum/median/maximum): 30/37.0/50
   Simulation truth: +0.531 pps, p=2.487e-14
============================================================================
==
=== SELECTION-THRESHOLD SENSITIVITY (20% MVC) ===
============================================================================
==
Simulation truth (all active MUs): +1.03 pps  (Normal 13.00, DPN 14.04)
Fixed: fmin=5 pps, fmax=15 pps, mn_number=10, 50 paired subjects
Common ordered seed set: 20261000-20261999 at every threshold

 ISI-CoV   median difference             95% seed range  min pool
    0.15              -0.607      -0.724 to  -0.486        19
    0.20              -0.601      -0.750 to  -0.449        23
    0.25              -0.586      -0.787 to  -0.377        27
    0.30              -0.495      -0.743 to  -0.267        32   <- main anal
ysis
    0.32              -0.440      -0.681 to  -0.187        34
    0.34              -0.364      -0.643 to  -0.107        34
    0.36              -0.311      -0.604 to  -0.040        34
    0.38              -0.203      -0.486 to  +0.076        34
    0.40              -0.149      -0.442 to  +0.135        35
    0.45              +0.048      -0.258 to  +0.350        35
    0.50              +0.125      -0.175 to  +0.429        35
```




```
Figure saved to: diabetes/figures/selection_threshold_sensitivity_v2.png
Sweep saved to: diabetes/csv_results/selection_threshold_sensitivity_v2.csv
```